

When More Thinking Is Not More Tokens

DataForge 2026 — latent recurrent reasoning

Modern reasoning models often make “thinking longer” visible as more generated tokens. A different idea is to spend more inference compute without producing a longer textual chain of thought: repeatedly update an internal state and only decode an answer after several latent computation steps.

Falsifiable idea: if a model has a useful recurrent computation, increasing its recurrence depth should sometimes improve reasoning on tasks that need more computation—but sufficiently large depth should not be assumed to help indefinitely.

Geiping et al. (2025) make this idea concrete with a recurrent-depth language model. Instead of generating more reasoning tokens, the model repeatedly unrolls a recurrent block at inference time. Their proof-of-concept reached 3.5 billion parameters and was trained on 800 billion tokens; they report that increasing test-time computation can improve reasoning performance, in some cases reaching a computation load described as equivalent to 50 billion parameters. [1] The important mechanism is simple: **the same learned computation is applied repeatedly, so inference gets an adjustable depth budget without requiring the intermediate computation to be written as language.**

That changes what “thinking longer” means. In token-based reasoning, additional computation usually creates additional observable sequence positions. In latent recurrence, additional computation can happen inside a fixed-dimensional state. The distinction is not merely cosmetic: a continuous internal state can represent intermediate information that would be awkward or expensive to serialize as natural language.

But recurrence depth is not a free accuracy knob. Kohli et al. (2026) study recurrent-depth Transformers on controlled systematic-generalization and depth-extrapolation problems. They report that scaling inference-time recurrence can unlock deeper compositions, but they also identify **overthinking**: excessive recurrence can degrade predictions. [2] That result makes the strongest version of the claim above intentionally modest. Recurrence can provide a compute budget; it does not guarantee that every extra unit of compute is useful.

Our interactive artifact turns that mechanism into a small experiment. The learner selects a synthetic reasoning problem, chooses the number of recurrent updates, runs the toy computation, and observes the numerical state trajectory and answer. The model is deliberately tiny. Its recurrence is an educational construction rather than the architecture from either paper. Its confidence score is also a teaching heuristic, not a calibrated probability. The point is to make the variable—**depth**—directly manipulable.

This is also where BDH-CQ becomes relevant. Engdahl et al. (2026) describe BDH-CQ as combining in-context learning with recurrent latent reasoning: inference-time inputs update recurrent memory, after which a query is solved through iterative computation in a high-dimensional latent space without verbalizing intermediate reasoning. Their reported 150M-parameter configuration reaches 29.5% pass@2 on ARC-AGI-1 at a computed inference cost of \$0.0007 per task. [3] Those numbers are properties of BDH-CQ, not of our toy.

The broader BDH architecture provides another useful connection. Kosowski et al. (2025) introduce Dragon Hatchling as a brain-inspired, sparse architecture built around locally interacting neuron particles and evolving internal state. The paper argues for a model in which memory and computation are more tightly coupled than in a conventional Transformer. [4] BDH-CQ applies the

recurrent-latent-reasoning idea within that architectural family.

There is an important observability limit, however. A visualization of latent dimensions is not automatically a visualization of semantic thoughts. Lu et al. (2025) probe a depth-recurrent Transformer and report limited evidence for an interpretable latent chain of thought. [5] Our artifact therefore shows the state because it is useful for teaching recurrence, not because each bar has a human-readable meaning.

What did not work? The first version of this educational model made it tempting to interpret higher confidence as genuine model certainty and the state bars as hidden reasoning steps. Neither interpretation is justified. We removed that implication by explicitly labeling the confidence as heuristic and the state as numerical.

My judgment: latent recurrence is most useful pedagogically when presented as a change in the *location and controllability of computation*, not as “secret chain-of-thought.” The interesting research question is whether repeated latent computation can become reliably more useful with depth while remaining efficient, trainable, and interpretable. Current evidence says that this is possible in controlled settings—but also that more recurrence can become counterproductive. That tension is the part worth learning.

References: [1] Geiping et al., 2025, <https://arxiv.org/abs/2502.05171>; [2] Kohli et al., 2026, <https://arxiv.org/abs/2604.07822>; [3] Engdahl et al., 2026, <https://arxiv.org/abs/2608.09888>; [4] Kosowski et al., 2025, <https://arxiv.org/abs/2509.26507>; [5] Lu et al., 2025, <https://arxiv.org/abs/2507.02199>.